

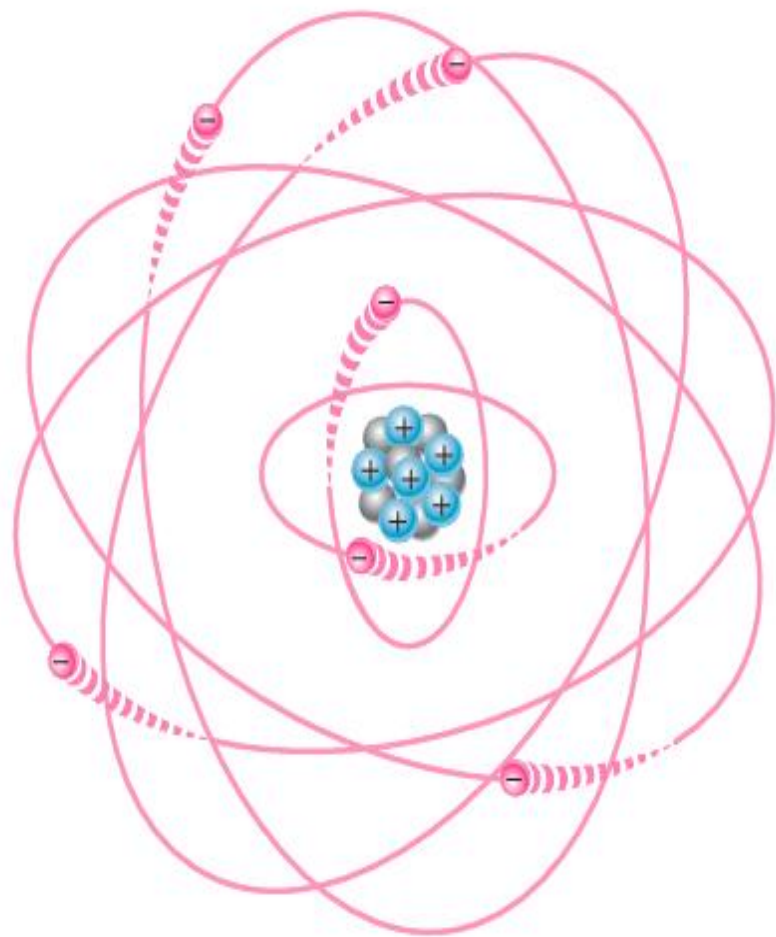
THE ATOM

- All matter is made up of atoms.
- Every atom contains electrons, protons, and neutrons.
- The only exception is normal hydrogen, which has no neutrons.
- Each element in the periodic table has a unique atomic structure.
- All atoms of the same element have the same number of protons.
- Early scientists thought atoms were tiny, indivisible solid spheres.
- Later, it was discovered that atoms have a small, dense nucleus.

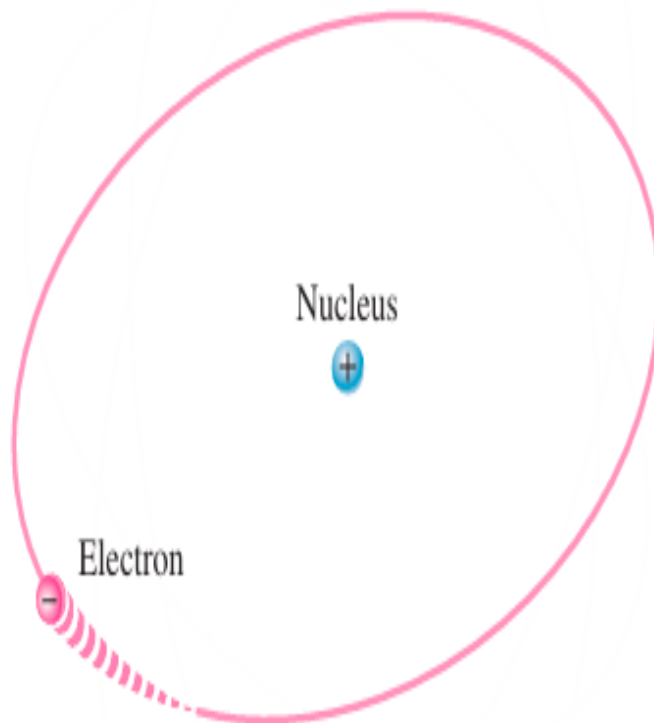
- Electrons orbit the nucleus at relatively large distances.
- Niels Bohr proposed that electrons move in fixed orbits around the nucleus.
- This concept is known as the Bohr (planetary) model of the atom.
- Another model called the quantum model is more accurate but harder to visualize.
- For most electronics applications, the Bohr model is used because it is simpler and easier to understand.

The Bohr Model

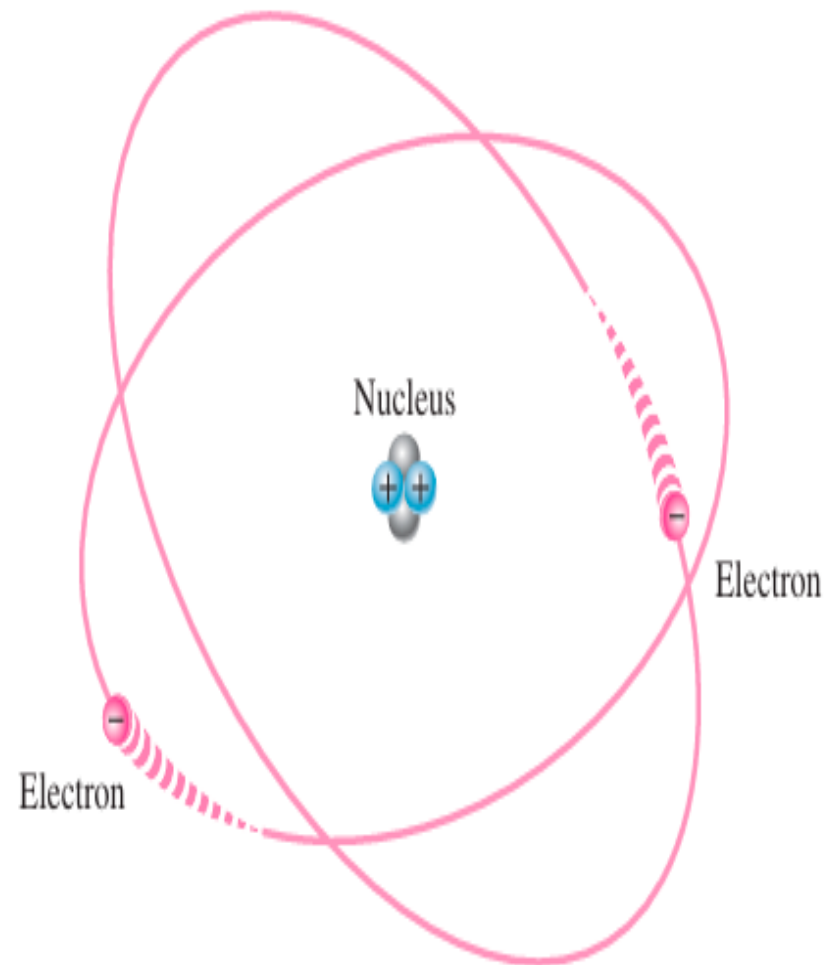
- An **atom** is the smallest particle of an element that retains the properties of that element.
- There are **118 known elements**, and each element has a unique type of atom.
- Atoms of one element are **different from atoms of other elements**.
- According to the **Bohr (planetary) model**, an atom has a central nucleus with electrons orbiting around it.
- The **nucleus** contains:
 - **Protons** (positively charged)
 - **Neutrons** (no charge)
- **Electrons** are negatively charged particles that move around the nucleus.
- Each element is identified by its **specific number of protons and electrons**.
- **Hydrogen** is the simplest atom, having:
 - 1 proton
 - 1 electron



⊖ Electron ⊕ Proton ● Neutron



(a) Hydrogen atom

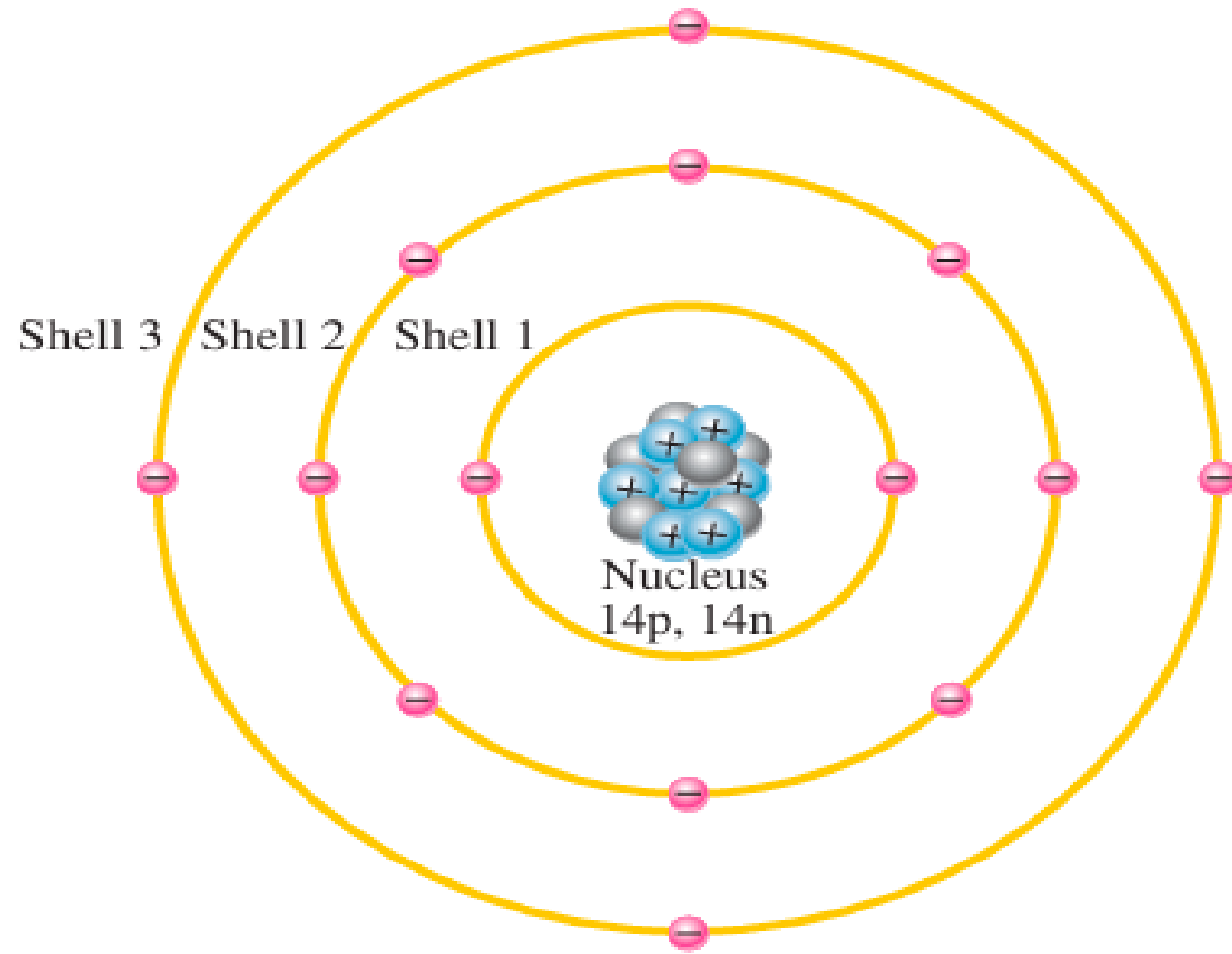


(b) Helium atom

Electrons and Shells Energy Levels

- Electrons move around the nucleus at fixed distances called orbits.
- Electrons closer to the nucleus have less energy.
- Electrons farther from the nucleus have more energy.
- Only discrete (specific) energy values are allowed for electrons in an atom.
- Because of this, electrons can move only in fixed orbits, not in between them.
- Each orbit has a specific energy level.
- The orbits are grouped into shells (energy levels).
- Each atom has a fixed number of shells.
- Each shell can hold only a maximum number of electrons.
- Shells are numbered 1, 2, 3, ..., with shell 1 being closest to the nucleus.
- Silicon atom:
 - There are 14 electrons around the nucleus
 - There are 14 protons and 14 neutrons in the nucleus

silicon atom.



The Maximum Number of Electrons in Each Shell The maximum number of electrons (N_e) that can exist in each shell of an atom is a fact of nature and can be calculated by the formula,

$$N_e = 2n^2$$

Valence Electrons

Electrons that are in orbits farther from the nucleus have higher energy and are less tightly bound to the atom than those closer to the nucleus. This is because the force of attraction between the positively charged nucleus and the negatively charged electron decreases with increasing distance from the nucleus. Electrons with the highest energy exist in the outermost shell of an atom and are relatively loosely bound to the atom. This outermost shell is known as the valence shell and electrons in this shell are called valence electrons. These valence electrons contribute to chemical reactions and bonding within the structure of a material and determine its electrical properties. When a valence electron gains sufficient energy from an external source, it can break free from its atom. This is the basis for conduction in materials.

Atoms are extremely small and cannot be seen even with the strongest optical microscopes; however, a scanning tunneling microscope can detect a single atom. The nucleus is so small and the electrons orbit at such distances that the atom is mostly empty space. To put it in perspective, if the proton in a hydrogen atom were the size of a golf ball, the electron orbit would be approximately one mile away.

Protons and neutrons are approximately the same mass. The mass of an electron is $1/1836$ of a proton. Within protons and neutrons there are even smaller particles called quarks.

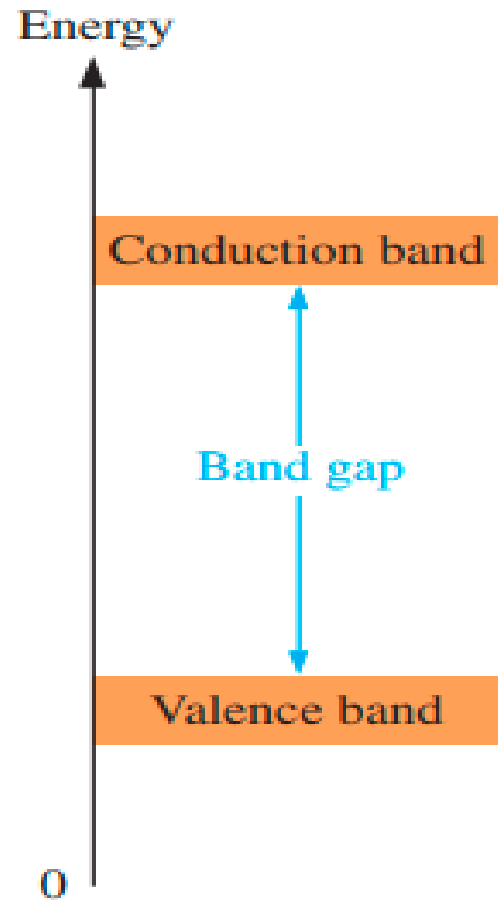
Valence Electrons

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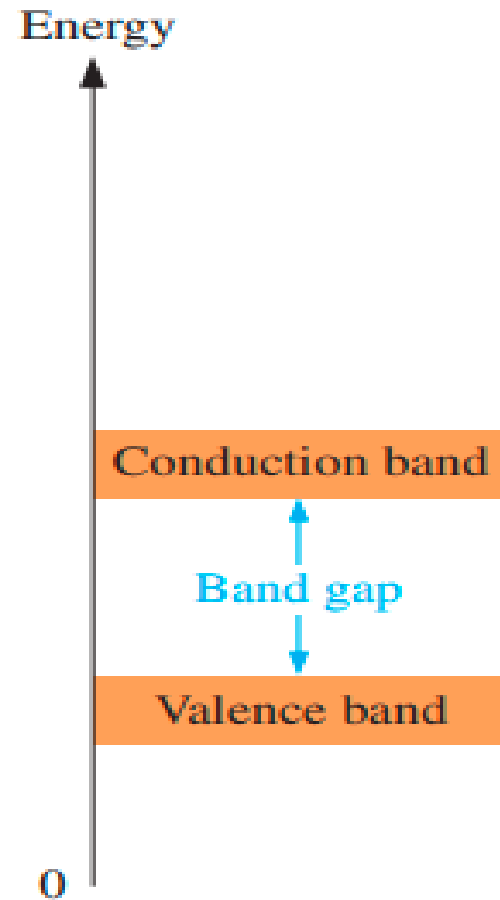
Band Gap

- *The difference in energy between the valence band and the conduction band is called an energy gap or band gap. This is the amount of energy that a valence electron must have in order to jump from the valence band to the conduction band. Once in the conduction band, the electron is free to move throughout the material and is not tied to any given atom.*
- *It is a region in insulators and semiconductors where no electron states exist. Although an electron may not exist in this region, it can “jump” across it under certain conditions. For insulators, the gap can be crossed only when breakdown conditions occur—as when a very high voltage is applied across the material.*
- *In semiconductors the band gap is smaller, allowing an electron in the valence band to jump into the conduction band if it absorbs a photon. The band gap depends on the semiconductor material.*
- *In conductors, the conduction band and valence band overlap, so there is no gap. This means that electrons in the valence band move freely into the conduction band, so there are always electrons available as free electrons.*

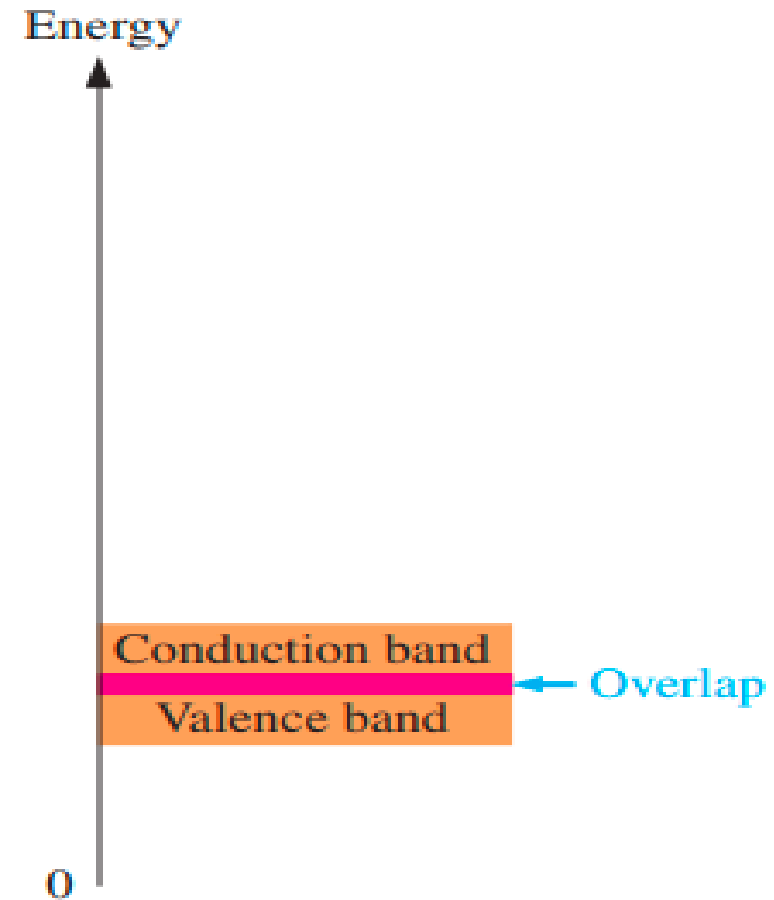
Energy diagrams for the three types of materials.



(a) Insulator



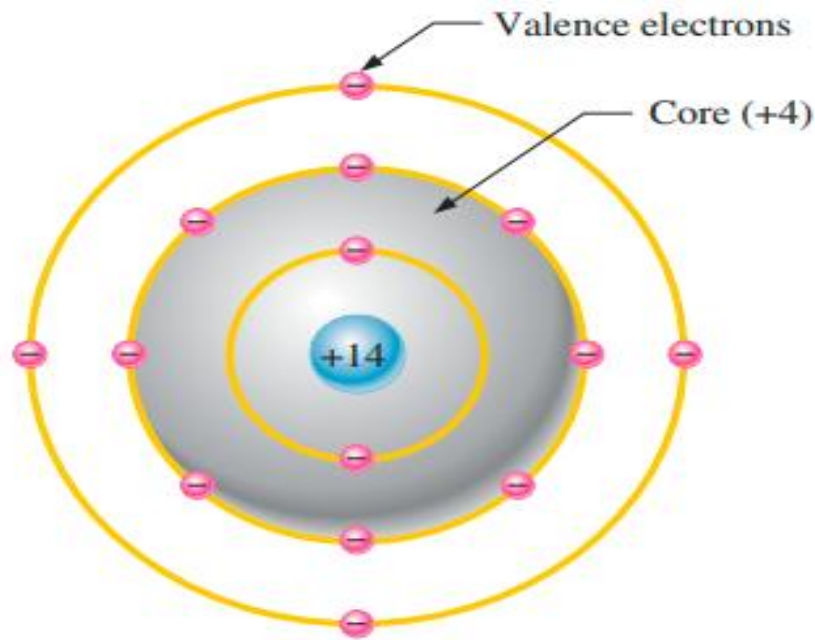
(b) Semiconductor



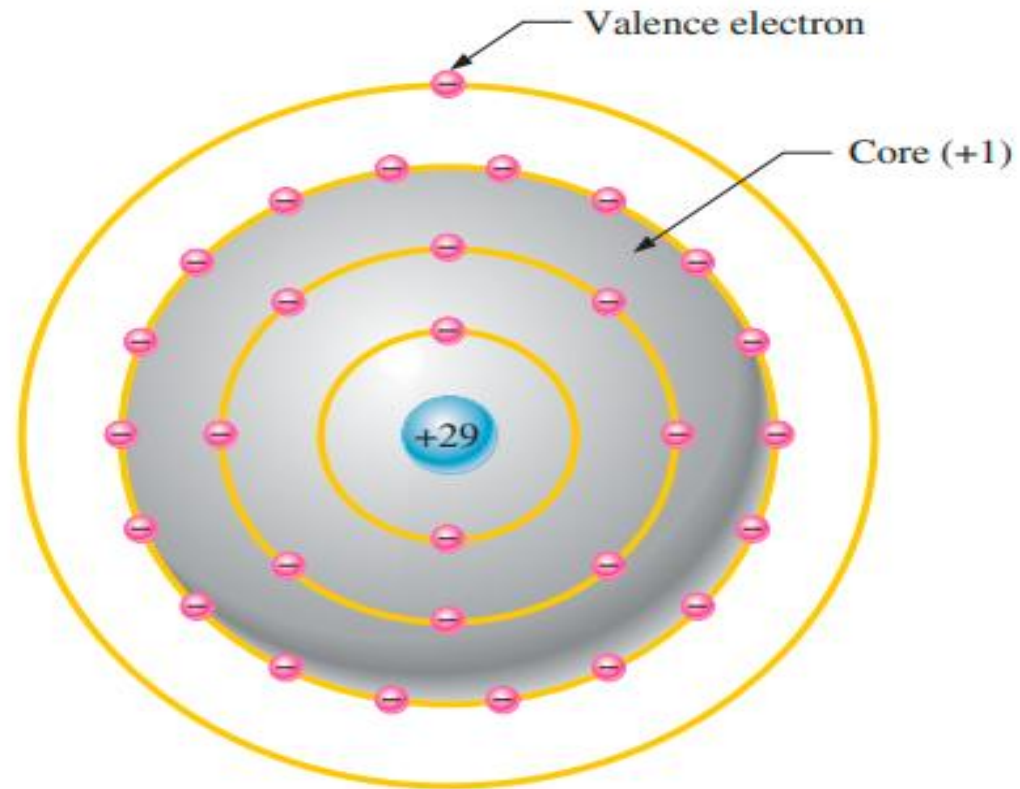
(c) Conductor

Comparison of a Semiconductor Atom to a Conductor Atom

- Silicon is a semiconductor and copper is a conductor. Bohr diagrams of the silicon atom and the copper atom. Notice that the core of the silicon atom has a net charge of 4 (14 protons 10 electrons) and the core of the copper atom has a net charge of 1 (29 protons 28 electrons). *The core includes everything except the valence electrons.*



(a) Silicon atom

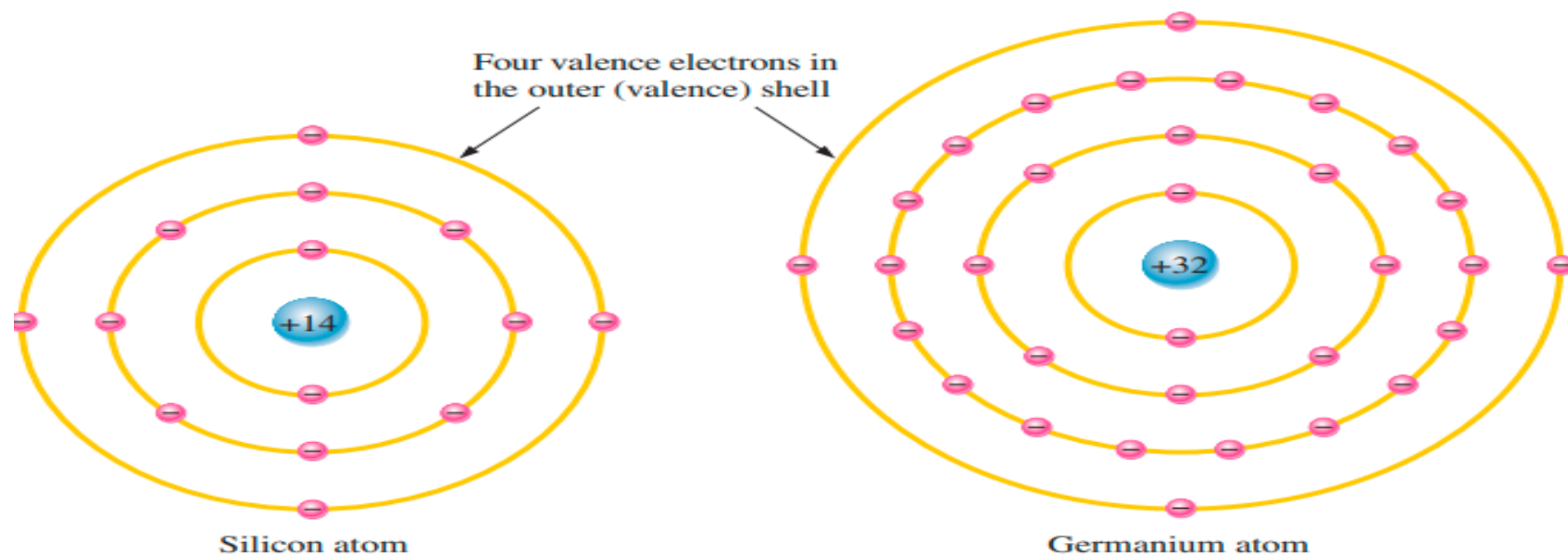


(b) Copper atom

- *The valence electron in the copper atom “feels” an attractive force of 1 compared to a valence electron in the silicon atom which “feels” an attractive force of 4. Therefore, there is more force trying to hold a valence electron to the atom in silicon than in copper.*
- *The copper’s valence electron is in the fourth shell, which is a greater distance from its nucleus than the silicon’s valence electron in the third shell. Recall that electrons farthest from the nucleus have the most energy. The valence electron in copper has more energy than the valence electron in silicon.*
- *This means that it is easier for valence electrons in copper to acquire enough additional energy to escape from their atoms and become free electrons than it is in silicon. In fact, large numbers of valence electrons in copper already have sufficient energy to be free electrons at normal room temperature.*

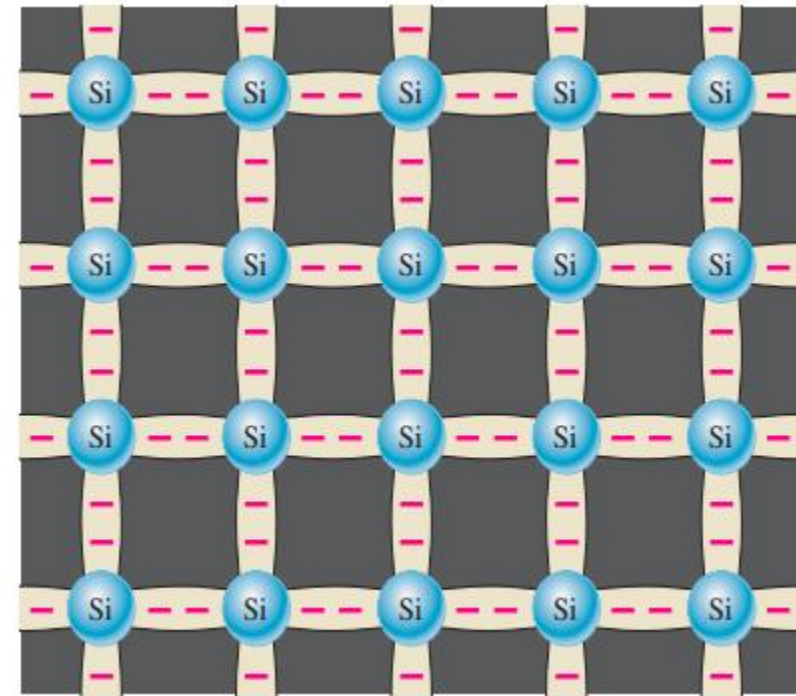
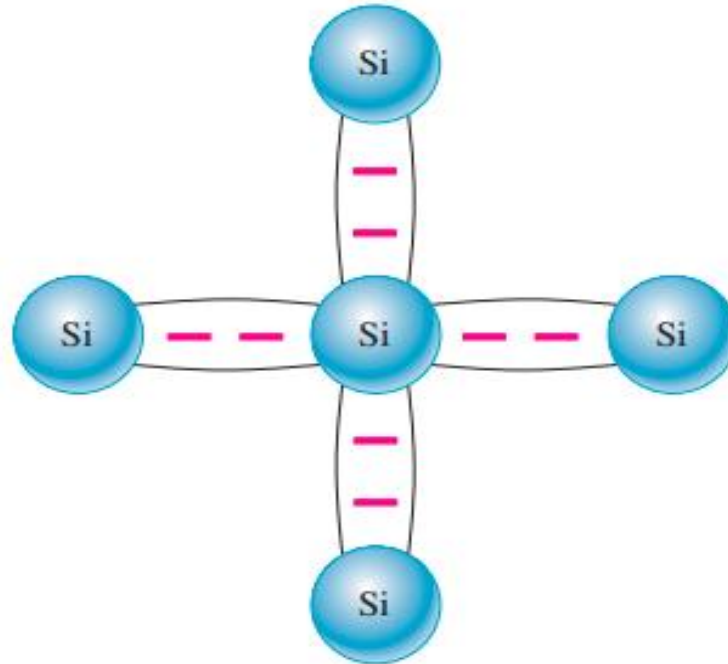
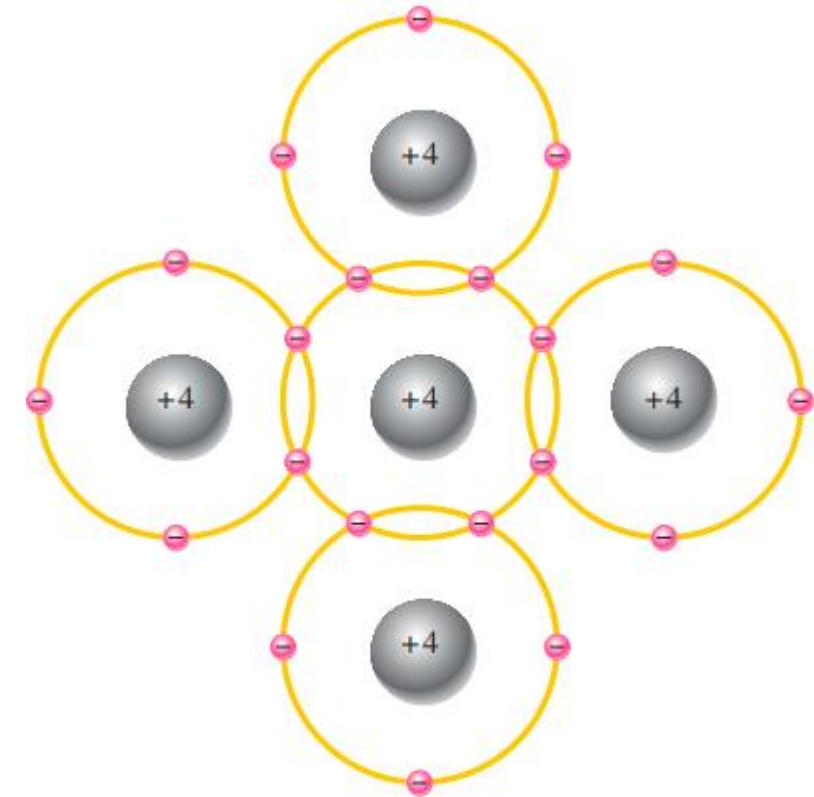
Silicon and Germanium

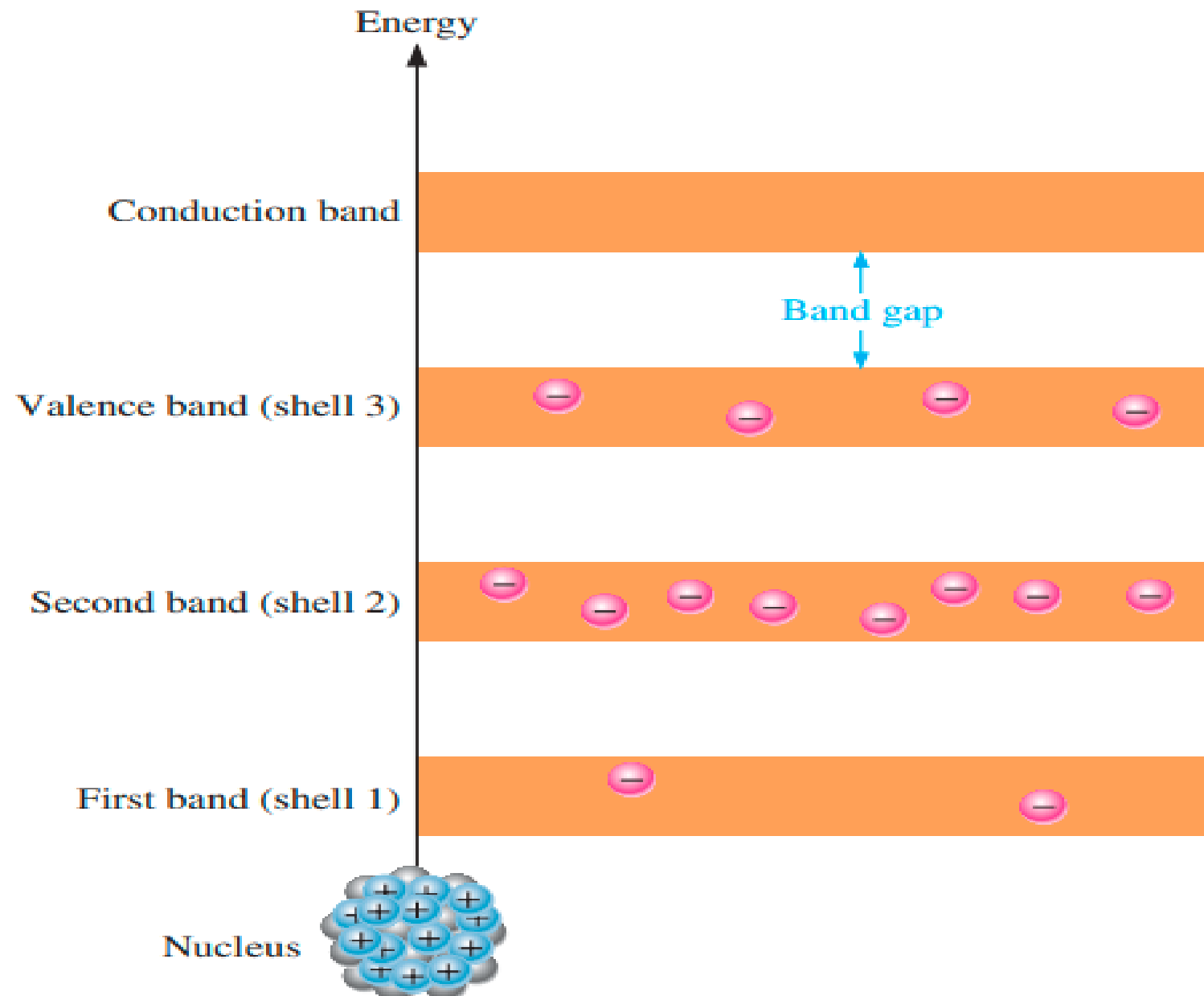
Silicon is used in diodes, transistors, integrated circuits, and other semiconductor devices. Notice that both silicon and germanium have the characteristic four valence electrons. The valence electrons in germanium are in the fourth shell while those in silicon are in the third shell, closer to the nucleus. This means that the germanium valence electrons are at higher energy levels than those in silicon and, therefore, require a smaller amount of additional energy to escape from the atom. This property makes germanium more unstable at high temperatures and results in excessive reverse current. This is why silicon is a more widely used semiconductive material.



Covalent Bonds

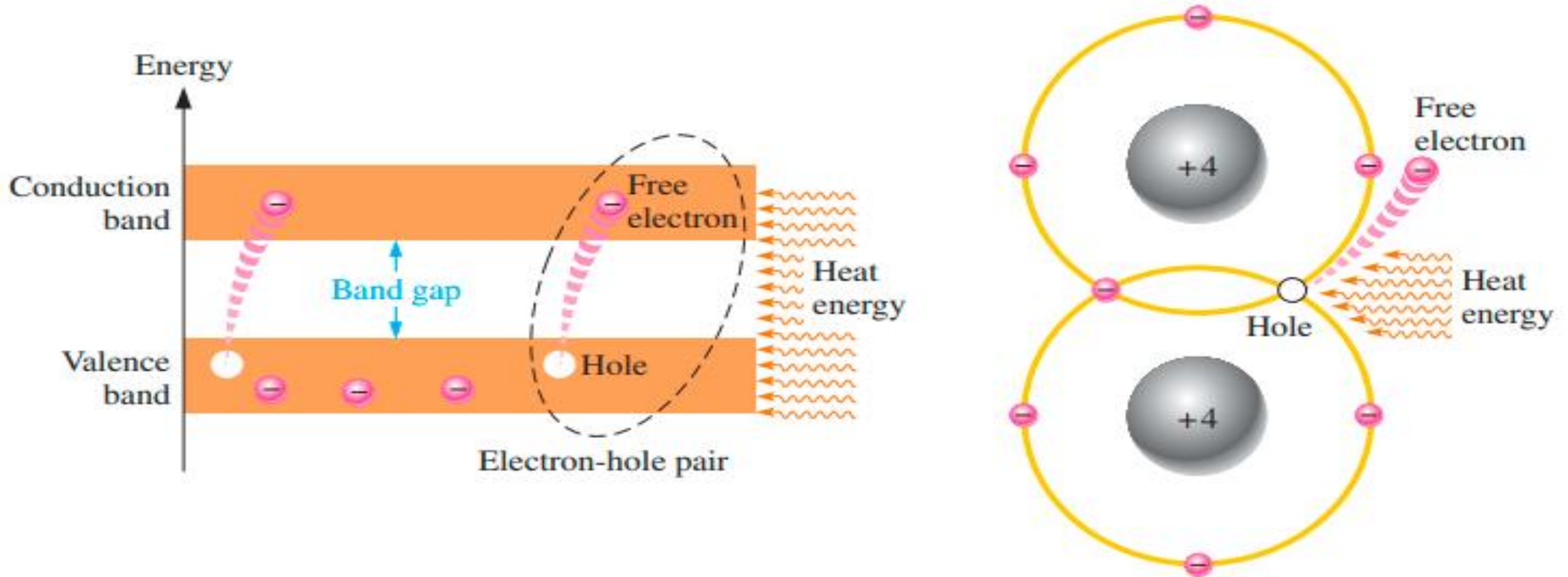
A silicon (Si) atom with its four valence electrons shares an electron with each of its four neighbors. This effectively creates eight shared valence electrons for each atom and produces a state of chemical stability. Also, this sharing of valence electrons produces the covalent bonds that hold the atoms together; each valence electron is attracted equally by the two adjacent atoms which share it. An intrinsic crystal is one that has no impurities. Covalent bonding for germanium is similar because it also has four valence electron.



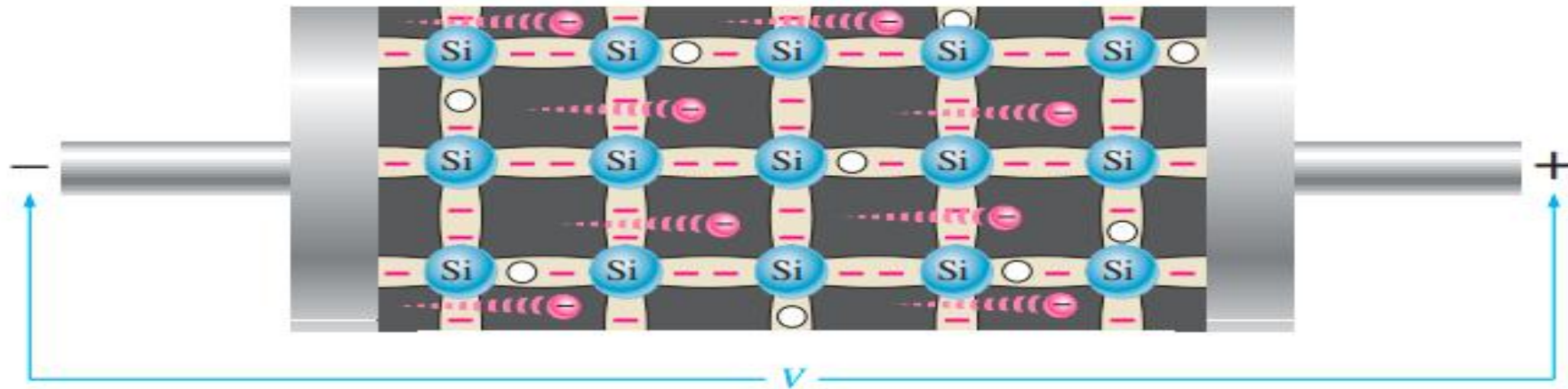


Conduction Electrons and Holes

- An intrinsic (pure) silicon crystal at room temperature has sufficient heat (thermal) energy for some valence electrons to jump the gap from the valence band into the conduction band, becoming free electrons. Free electrons are also called conduction electrons.*



- *When an electron jumps to the conduction band, a vacancy is left in the valence band within the crystal. This vacancy is called a hole. For every electron raised to the conduction band by external energy, there is one hole left in the valence band, creating what is called an **electron-hole pair**. Recombination occurs when a conduction-band electron loses energy and falls back into a hole in the valence band.*
- *When a voltage is applied across a piece of intrinsic silicon the thermally generated free electrons in the conduction band, which are free to move randomly in the crystal structure, are now easily attracted toward the positive end. This movement of free electrons is one type of current in a semiconductive material and is called electron current.*



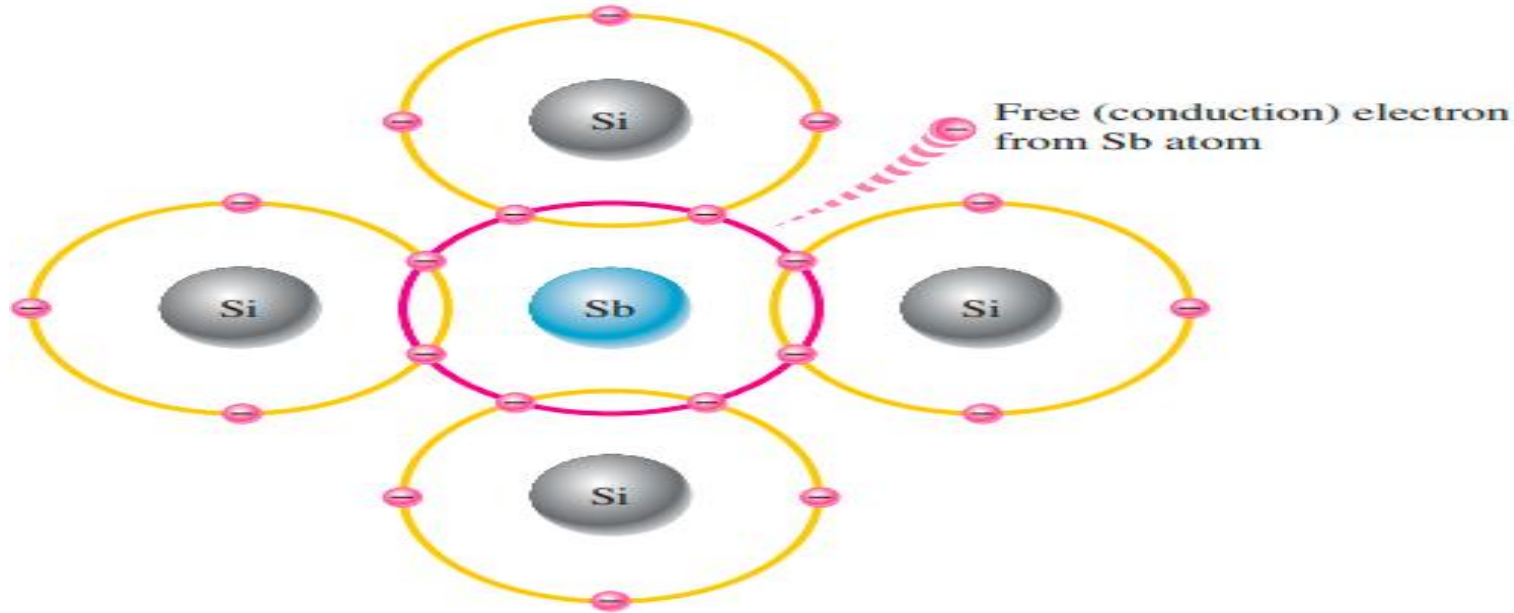
- *Another type of current occurs in the valence band, where the holes created by the free electrons exist. Electrons remaining in the valence band are still attached to their atoms and are not free to move randomly in the crystal structure as are the free electrons. However, a valence electron can move into a nearby hole with little change in its energy level, thus leaving another hole where it came from.*
- *Although current in the valence band is produced by valence electrons, it is called hole current to distinguish it from electron current in the conduction band*

Semiconductive materials do not conduct current well and are of limited value in their intrinsic state. This is because of the limited number of free electrons in the conduction band and holes in the valence band. Intrinsic silicon (or germanium) must be modified by increasing the number of free electrons or holes to increase its conductivity and make it useful in electronic devices. This is done by adding impurities to the intrinsic material. Two types of extrinsic (impure) semiconductive materials, n -type and p -type, are the key building blocks for most types of electronic devices.

Since semiconductors are generally poor conductors, their conductivity can be drastically increased by the controlled addition of impurities to the intrinsic (pure) semiconductive material. This process, called **doping**, increases the number of current carriers (electrons or holes). The two categories of impurities are *n*-type and *p*-type.

N-Type Semiconductor

- To increase the number of conduction-band electrons in intrinsic silicon, pentavalent impurity atoms are added. These are atoms with five valence electrons such as arsenic (As), phosphorus (P), bismuth (Bi), and antimony (Sb).*



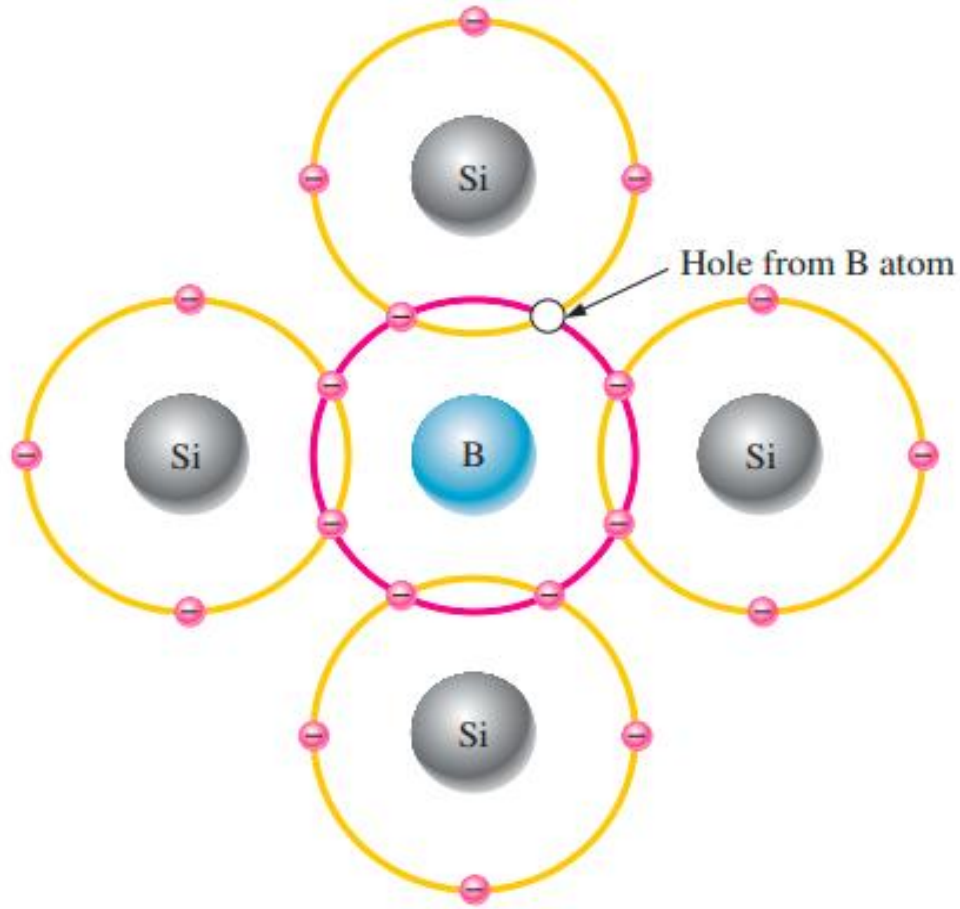
Each pentavalent atom (antimony, in this case) forms covalent bonds with four adjacent silicon atoms. Four of the antimony atom's valence electrons are used to form the covalent bonds with silicon atoms, leaving one extra electron. This extra electron becomes a conduction electron because it is not involved in bonding. Because the pentavalent atom gives up an electron, it is often called a donor atom. The number of conduction electrons can be carefully controlled by the number of impurity atoms added to the silicon. A conduction electron created by this doping process does not leave a hole in the valence band because it is in excess of the number required to fill the valence band.

Majority and Minority Carriers

- *Since most of the current carriers are electrons, silicon (or germanium) doped with pentavalent atoms is an N-type semiconductor (the n stands for the negative charge on an electron). The electrons are called the majority carriers in n-type material. Although the majority of current carriers in n-type material are electrons, there are also a few holes that are created when electron-hole pairs are thermally generated. These holes are not produced by the addition of the pentavalent impurity atoms. Holes in an n-type material are called minority carriers.*

P-Type Semiconductor

- To increase the number of holes in intrinsic silicon, trivalent impurity atoms are added. These are atoms with three valence electrons such as boron (B), indium (In), and gallium (Ga). Each trivalent atom (boron, in this case) forms covalent bonds with four adjacent silicon atoms.
- All three of the boron atom's valence electrons are used in the covalent bonds; and, since four electrons are required, a hole results when each trivalent atom is added. Because the trivalent atom can take an electron, it is often referred to as an acceptor atom. The number of holes can be carefully controlled by the number of trivalent impurity atoms added to the silicon. A hole created by this doping process is not accompanied by a conduction (free) electron.



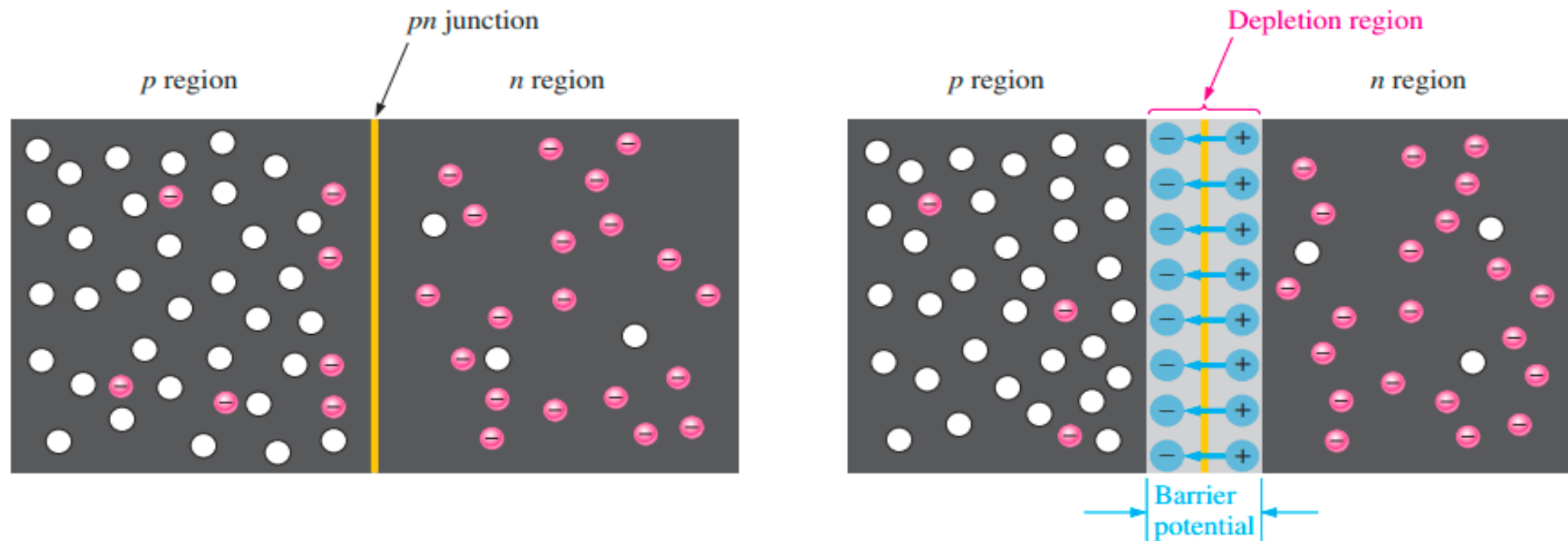
Majority and Minority Carriers Since most of the current carriers are holes, silicon (or germanium) doped with trivalent atoms is called a p-type semiconductor. The holes are the majority carriers in p-type material. Although the majority of current carriers in p-type material are holes, there are also a few conduction-band electrons that are created when electron-hole pairs are thermally generated. These conduction-band electrons are not produced by the addition of the trivalent impurity atoms. Conduction-band electrons in p-type material are the minority carriers.

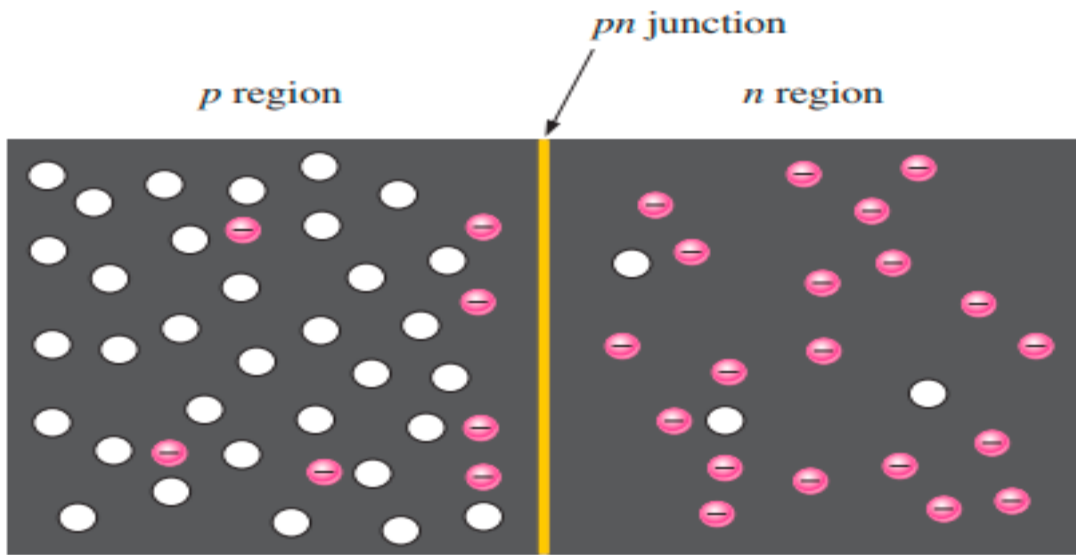
When you take a block of silicon and dope part of it with a trivalent impurity and the other part with a pentavalent impurity, a boundary called the *pn* junction is formed between the resulting *p*-type and *n*-type portions. The *pn* junction is the basis for diodes, certain transistors, solar cells, and other devices, as you will learn later.

A P-type material consists of silicon atoms and trivalent impurity atoms such as boron. The boron atom adds a hole when it bonds with the silicon atoms. However, since the number of protons and the number of electrons are equal throughout the material, there is no net charge in the material and so it is neutral.

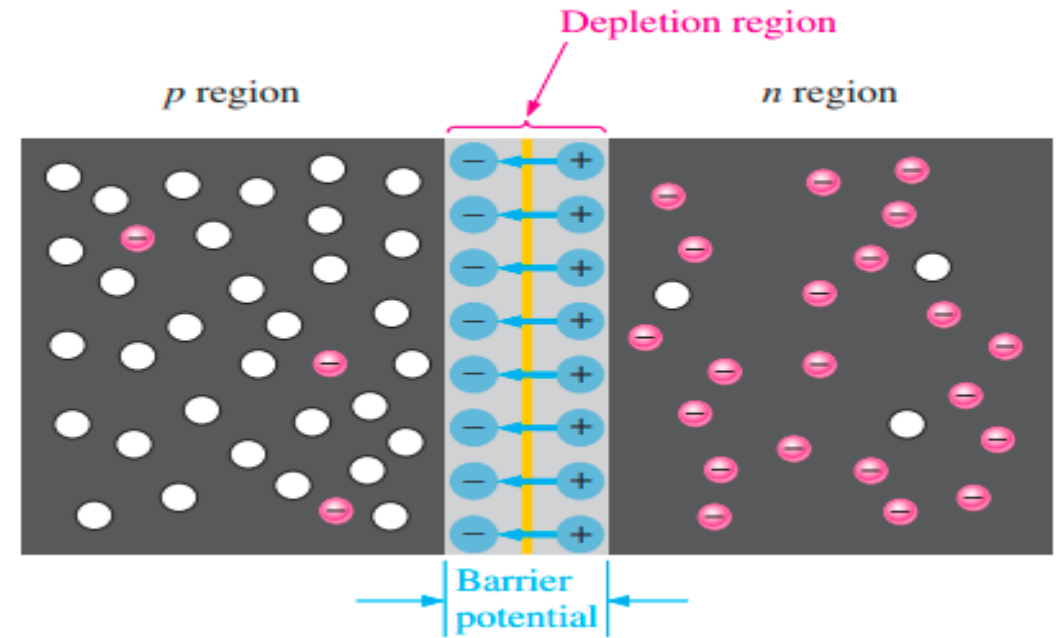
An N-type silicon material consists of silicon atoms and pentavalent impurity atoms such as antimony. As you have seen, an impurity atom releases an electron when it bonds with four silicon atoms. Since there is still an equal number of protons and electrons (including the free electrons) throughout the material, there is no net charge in the material and so it is neutral.

- If a piece of intrinsic silicon is doped so that part is N-type and the other part is P-type, a P.N junction forms at the boundary between the two regions and a diode is created. The p region has many holes (majority carriers) from the impurity atoms and only a few thermally generated free electrons (minority carriers). The N region has many free electrons (majority carriers) from the impurity atoms and only a few thermally generated holes (minority carriers).*





The basic silicon structure at the instant of junction formation showing only the majority and minority carriers. Free electrons in the N region near the PN junction begin to diffuse across the junction and fall into holes near the junction in the P region.



For every electron that diffuses across the junction and combines with a hole, a positive charge is left in the *n* region and a negative charge is created in the *p* region, forming a barrier potential. This action continues until the voltage of the barrier repels further diffusion. The blue arrows between the positive and negative charges in the depletion region represent the electric field.

Formation of the Depletion Region

The free electrons in the N region are randomly drifting in all directions. At the instant of the PN junction formation, the free electrons near the junction in the N region begin to diffuse across the junction into the P region where they combine with holes near the junction.

Before the PN junction is formed, recall that there are as many electrons as protons in the N-type material, making the material neutral in terms of net charge. The same is true for the p-type material. When the PN junction is formed, the N region loses free electrons as they diffuse across the junction.

This creates a layer of positive charges (pentavalent ions) near the junction. As the electrons move across the junction, the P region loses holes as the electrons and holes combine. This creates a layer of negative charges (trivalent ions) near the junction. These two layers of positive and negative charges form the depletion region.

The term depletion refers to the fact that the region near the PN junction is depleted of charge carriers (electrons and holes) due to diffusion across the junction. Keep in mind that the depletion region is formed very quickly and is very thin compared to the N region and P region

In other words, the depletion region acts as a barrier to the further movement of electrons across the junction.

HISTORY NOTE

After the invention of the light bulb, Edison continued to experiment and in 1883 found that he could detect electrons flowing through the vacuum from the lighted filament to a metal plate mounted inside the bulb. This discovery became known as the *Edison effect*.

An English physicist, John Fleming, took up where Edison left off and found that the Edison effect could also be used to detect radio waves and convert them to electrical signals. He went on to develop a two-element vacuum tube called the *Fleming valve*, later known as the *diode*. Modern *pn* junction devices are an outgrowth of this.

HISTORY NOTE

Russell Ohl, working at Bell Labs in 1940, stumbled on the semiconductor *pn* junction. Ohl was working with a silicon sample that had an accidental crack down its middle. He was using an ohmmeter to test the electrical resistance of the sample when he noted that when the sample was exposed to light, the current that flowed between the two sides of the crack made a significant jump. This discovery was fundamental to the work of the team that invented the transistor in 1947.

Barrier Potential

In the depletion region there are many positive charges and many negative charges on opposite sides of the PN junction. The forces between the opposite charges form an electric field. This electric field is a barrier to the free electrons in the N region, and energy must be expended to move an electron through the electric field.

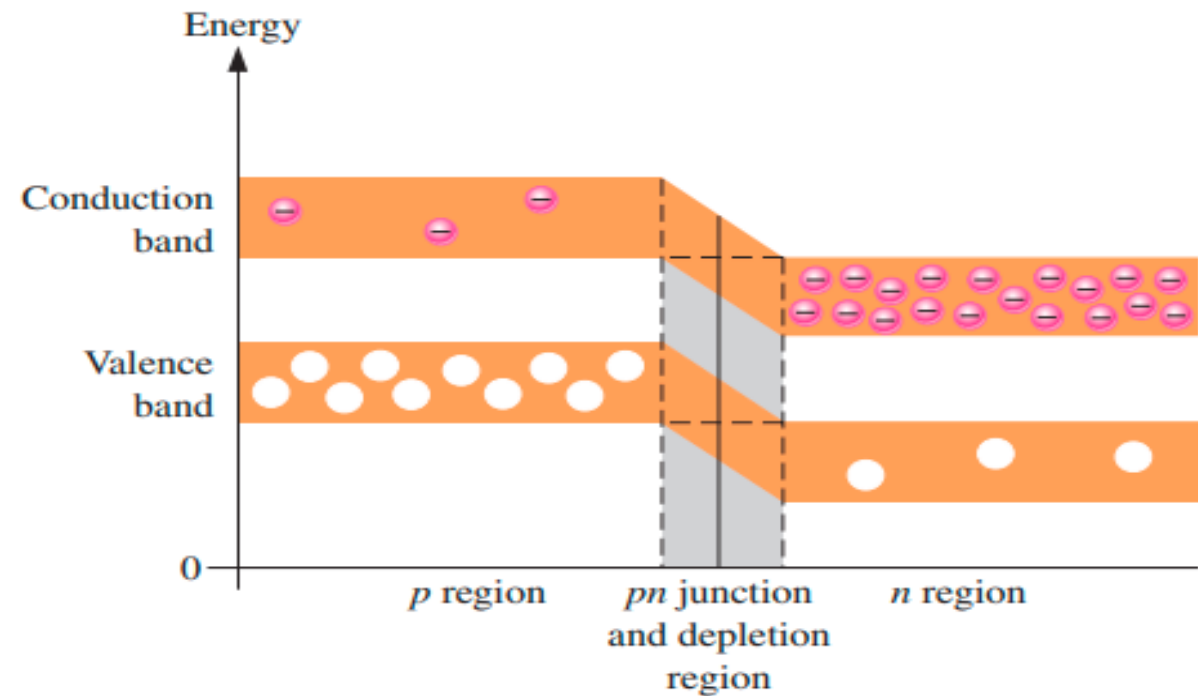
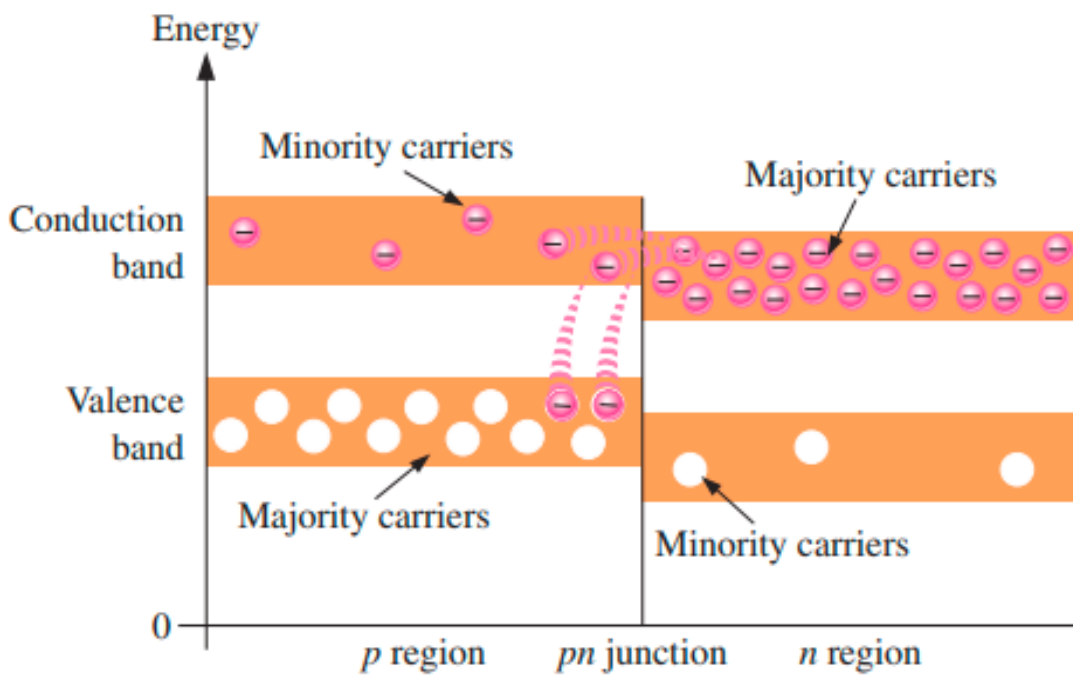
That is, external energy must be applied to get the electrons to move across the barrier of the electric field in the depletion region. The potential difference of the electric field across the depletion region is the amount of voltage required to move electrons through the electric field. This potential difference is called the barrier potential and is expressed in volts.

Stated another way, a certain amount of voltage equal to the barrier potential and with the proper polarity must be applied across a PN junction before electrons will begin to flow across the junction.

The barrier potential of a PN junction depends on several factors, including the type of semiconductive material, the amount of doping, and the temperature. The typical barrier potential is approximately 0.7 V for silicon and 0.3 V for germanium. Because germanium devices are not widely used, silicon will be used throughout the rest of the book.

Energy Diagrams of the PN Junction and Depletion Region

The valence and conduction bands in an N-type material are at slightly lower energy levels than the valence and conduction bands in a P-type material. Recall that P-type material has trivalent impurities and N-type material has pentavalent impurities. The trivalent impurities exert lower forces on the outer-shell electrons than the pentavalent impurities. The lower forces in P-type materials mean that the electron orbits are slightly larger and hence have greater energy than the electron orbits in the N-type materials.



As you can see, the valence and conduction bands in the n region are at lower energy levels than those in the p region, but there is a significant amount of overlapping.

SUMMARY

- ◆ According to the classical Bohr model, the atom is viewed as having a planetary-type structure with electrons orbiting at various distances around the central nucleus.
- ◆ According to the quantum model, electrons do not exist in precise circular orbits as particles as in the Bohr model. The electrons can be waves or particles and precise location at any time is uncertain.
- ◆ The nucleus of an atom consists of protons and neutrons. The protons have a positive charge and the neutrons are uncharged. The number of protons is the atomic number of the atom.
- ◆ Electrons have a negative charge and orbit around the nucleus at distances that depend on their energy level. An atom has discrete bands of energy called *shells* in which the electrons orbit. Atomic structure allows a certain maximum number of electrons in each shell. In their natural state, all atoms are neutral because they have an equal number of protons and electrons.
- ◆ The outermost shell or band of an atom is called the *valence band*, and electrons that orbit in this band are called *valence electrons*. These electrons have the highest energy of all those in the atom. If a valence electron acquires enough energy from an outside source such as heat, it can jump out of the valence band and break away from its atom.

- ◆ Insulating materials have very few free electrons and do not conduct current at all under normal circumstances.
- ◆ Materials that are conductors have a large number of free electrons and conduct current very well.
- ◆ Semiconductive materials fall in between conductors and insulators in their ability to conduct current.
- ◆ Semiconductor atoms have four valence electrons. Silicon is the most widely used semiconductive material.
- ◆ Semiconductor atoms bond together in a symmetrical pattern to form a solid material called a *crystal*. The bonds that hold a crystal together are called *covalent bonds*.
- ◆ The valence electrons that manage to escape from their parent atom are called *conduction electrons* or *free electrons*. They have more energy than the electrons in the valence band and are free to drift throughout the material.
- ◆ When an electron breaks away to become free, it leaves a hole in the valence band creating what is called an *electron-hole pair*. These electron-hole pairs are thermally produced because the electron has acquired enough energy from external heat to break away from its atom.
- ◆ A free electron will eventually lose energy and fall back into a hole. This is called *recombination*. Electron-hole pairs are continuously being thermally generated so there are always free electrons in the material.
- ◆ When a voltage is applied across the semiconductor, the thermally produced free electrons move toward the positive end and form the current. This is one type of current and is called electron current.
- ◆ Another type of current is the hole current. This occurs as valence electrons move from hole to hole creating, in effect, a movement of holes in the opposite direction.

- ◆ An n -type semiconductive material is created by adding impurity atoms that have five valence electrons. These impurities are *pentavalent atoms*. A p -type semiconductor is created by adding impurity atoms with only three valence electrons. These impurities are *trivalent atoms*.
- ◆ The process of adding pentavalent or trivalent impurities to a semiconductor is called *doping*.
- ◆ The majority carriers in an n -type semiconductor are free electrons acquired by the doping process, and the minority carriers are holes produced by thermally generated electron-hole pairs. The majority carriers in a p -type semiconductor are holes acquired by the doping process, and the minority carriers are free electrons produced by thermally generated electron-hole pairs.
- ◆ A pn junction is formed when part of a material is doped n -type and part of it is doped p -type. A depletion region forms starting at the junction that is devoid of any majority carriers. The depletion region is formed by ionization.
- ◆ The barrier potential is typically 0.7 V for a silicon diode and 0.3 V for germanium.

KEY TERMS

Atom The smallest particle of an element that possesses the unique characteristics of that element.

Barrier potential The amount of energy required to produce full conduction across the *pn* junction in forward bias.

Conductor A material that easily conducts electrical current.

Crystal A solid material in which the atoms are arranged in a symmetrical pattern.

Doping The process of imparting impurities to an intrinsic semiconductive material in order to control its conduction characteristics.

Electron The basic particle of negative electrical charge.

Free electron An electron that has acquired enough energy to break away from the valence band of the parent atom; also called a *conduction electron*.

Hole The absence of an electron in the valence band of an atom.

Insulator A material that does not normally conduct current.

Ionization The removal or addition of an electron from or to a neutral atom so that the resulting atom (called an ion) has a net positive or negative charge.

Orbital Subshell in the quantum model of an atom.

PN junction The boundary between two different types of semiconductive materials.

Proton The basic particle of positive charge.

Semiconductor A material that lies between conductors and insulators in its conductive properties. Silicon, germanium, and carbon are examples.

Shell An energy band in which electrons orbit the nucleus of an atom.

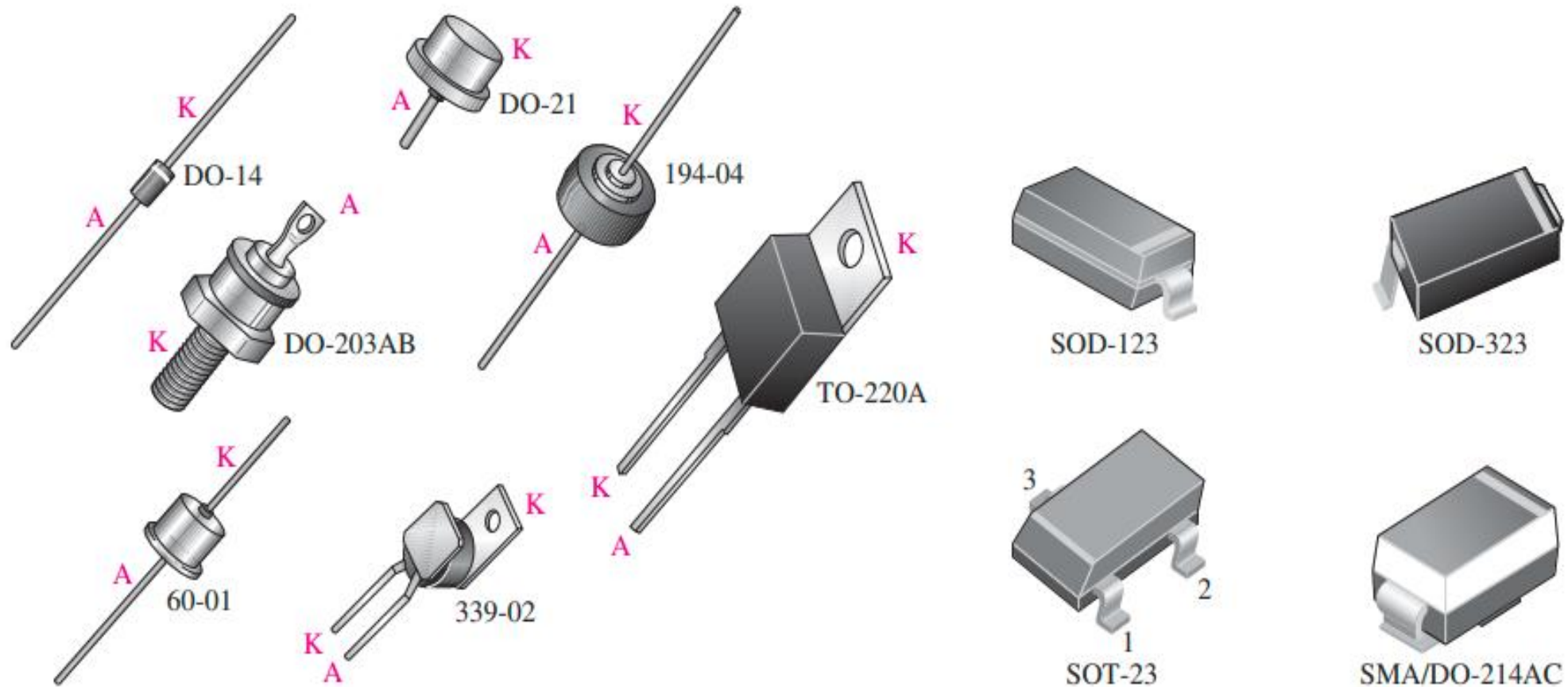
Silicon A semiconductive material.

Valence Related to the outer shell of an atom.

$$N_e = 2n^2$$

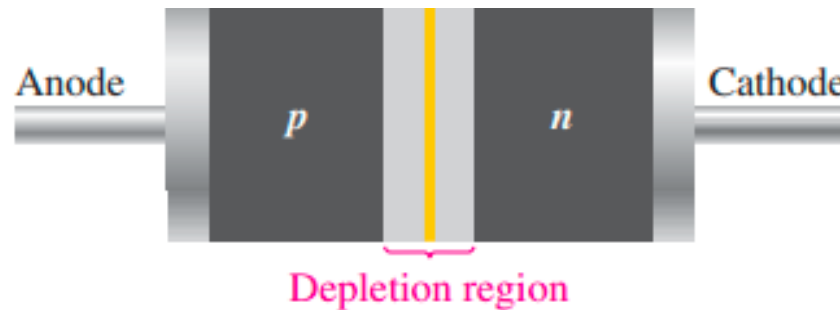
CHAPTER # 2

The Diodes



- A **diode** is a two-terminal semiconductor device formed by two doped regions of silicon separated by a PN junction. The most common category of diode, known as the general-purpose diode, is covered. Other names, such as rectifier diode or signal diode, depend on the particular type of application for which the diode was designed. You will learn how to use a voltage to cause the diode to conduct current in one direction and block it in the other direction. This process is called *biasing*.

A **diode** is made from a small piece of semiconductor material, usually silicon, in which half is doped as a P region and half is doped as an N region with a PN junction and depletion region in between. The P region is called the anode and is connected to a conductive terminal. The N region is called the cathode and is connected to a second conductive terminal.



(a) Basic structure



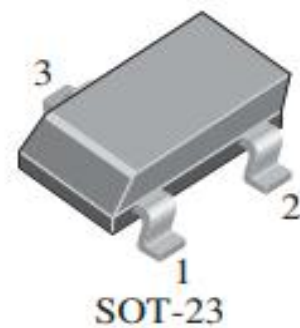
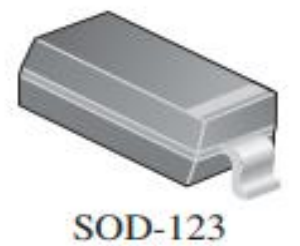
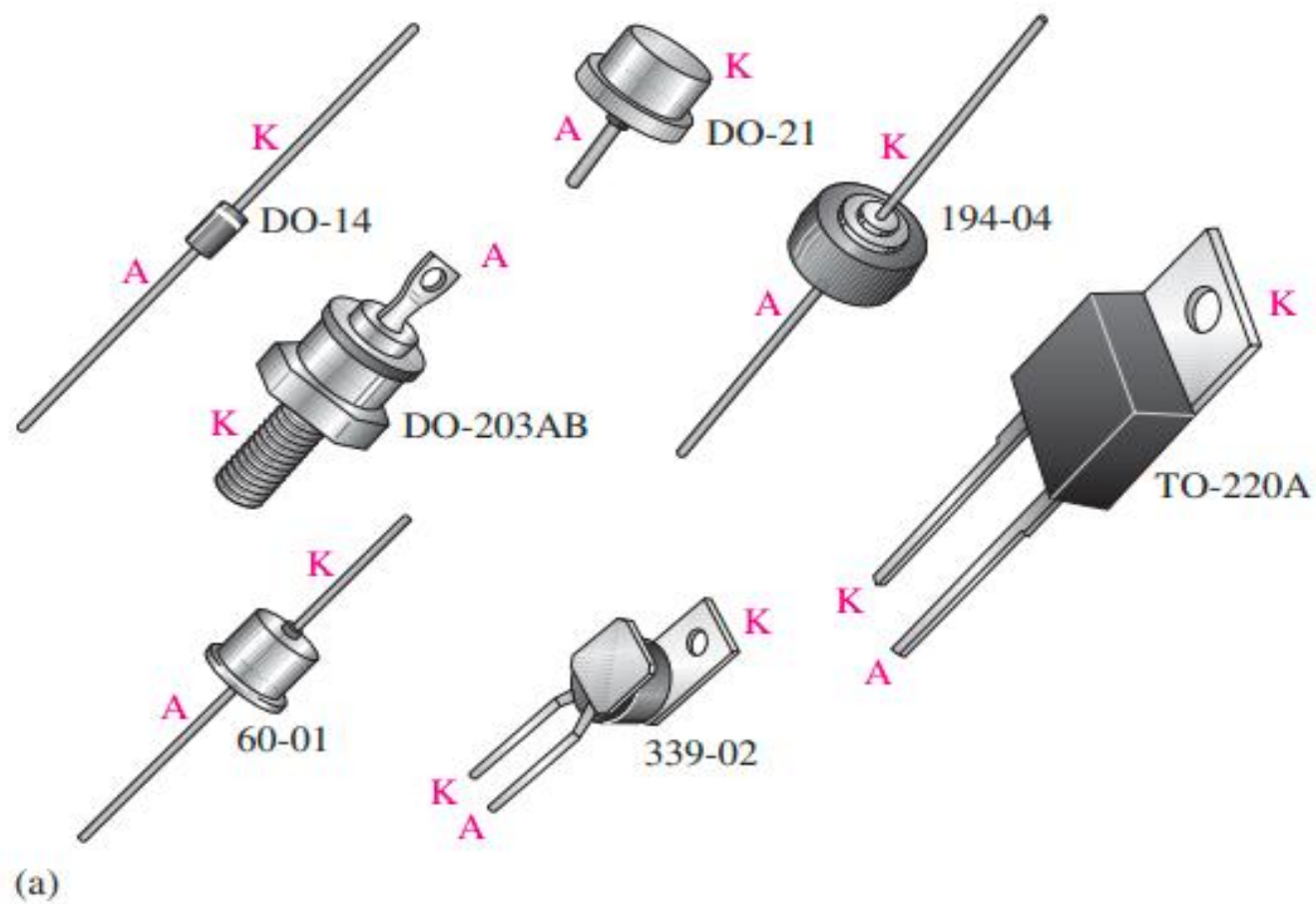
(b) Symbol

Typical Diode Packages

- *The anode (A) and cathode (K) are indicated on a diode in several ways, depending on the type of package. The cathode is usually marked by a band, a tab, or some other feature. On those packages where one lead is connected to the case, the case is the cathode.*

Surface-Mount Diode Packages

- *Typical diode packages for surface mounting on a printed circuit board. The SOD and SOT packages have gull-wing shaped leads. The SMA package has L-shaped leads that bend under the package. The SOD and SMA types have a band on one end to indicate the cathode. The SOT type is a three-terminal package in which there are either one or two diodes. In a single-diode SOT package, pin 1 is usually the anode and pin 3 is the cathode. In a dual-diode SOT package, pin 3 is the common terminal and can be either the anode or the cathode. Always check the datasheet for the particular diode to verify the pin configurations.*



Forward Bias

- To bias a diode, you apply a dc voltage across it. Forward bias is the condition that allows current through the PN junction. A dc voltage source connected by conductive material (contacts and wire) across a diode in the direction to produce forward bias. This external bias voltage is designated as V_{BIAS} . The resistor limits the forward current to a value that will not damage the diode. Notice that the negative side of V_{BIAS} is connected to the N region of the diode and the positive side is connected to the P region. This is one requirement for forward bias. A second requirement is that the bias voltage, V_{BIAS} , must be greater than the barrier potential.

